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Frame Semantics for EU Reporting Requirements

An Annotated Benchmark, Guidelines, and a Path Towards RRMV Population

Célestine Juliette Marie Lecat¹
Kimmy Shah² · Monica Palmirani¹

¹CIRSFID ALMA-AI, Department of Legal Studies University of Bologna, Bologna, Italy

²LSTS Research Group, Vrije Universiteit Brussel, Brussels, Belgium

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Some context

Reporting requirements in EU law

Provisions that oblige an institution, agency or Member State to supply a report, technical standard, opinion or notification to another EU body.

1

Growing volume

Number and complexity have risen sharply; institutions struggle with overlapping mandates.

2

Moving target

The underlying legislation keeps changing through amendments and derogations.

3

Still manual

Identification and monitoring remain largely manual legal review.

4

Policy pressure

EU regulatory simplification presupposes that these obligations can first be identified at scale.

91

reporting requests annotated

Source corpus

Corazza et al. (2025), developed within the SORTIS project - Study on Regulatory Reporting Standards.

Research questions

RQ1

Can frame semantics give a fine-grained, reproducible representation of the linguistic structure of reporting requests?

RQ2

Is the scheme reliable: can another annotator reproduce it from the guidelines alone?

RQ3

Can the layer be mapped onto an existing legal knowledge representation, and what does the attempt reveal?

Why an intermediate layer?

An LLM could be prompted to produce ontology instances directly

1

Independence from the target

The annotation records linguistic structure, not commitments to one vocabulary, so it survives revisions of the target model. RRMV is already being succeeded by LOMO.

2

Reuse across languages

Frames and frame elements are defined independently of any language, which matters for a body of law existing in 24 official languages.

3

Localisable evaluation

Frame-semantic parsing decomposes into three subtasks with existing parsers and benchmarks, so an error can be attributed to a stage rather than only observed as a wrong output.

Frame semantics

1

Frame

A structured description of a type of situation, with the participants relevant to it.

2

Lexical Unit

The word that evokes the frame. In "ESMA shall submit the standards", submit evokes/triggers *Telling*.

3

Frame Element

A participant slot once filled by a span of text: Speaker, Addressee, Message.

Worked example

Telling

ESMA [Speaker] shall submit those draft technical standards [Message] to the Commission [Addressee] by 2018-07-21.

Frame Elements

LU: submit.v → Frame: *Telling*

Disambiguation

Dealing with homonyms

"Appeal"

Disambiguation

In FrameNet

Lexical Unit Search

Q

Enter lexical unit name...

i

Search

There are 4 matching lexical units.

appeal.n

(from frame [Request](#))

appeal.n

(from frame [Appeal](#))

appeal.v

(from frame [Appeal](#))

appeal.v

(from frame [Stimulate_emotion](#))

Disambiguation

Anchored in lexicon

Lexical Unit = trigger

Appeal

The defense [Representative] plans to **appeal** the court's decision [Decision] on the grounds that the judge violated Code 81.b [Grounds].

Frame Elements

Request

His owner [Speaker] **briskly** [Manner] **appealed** to the sheepdog [Addressee] for help with the flock [Message].

Stimulate_emotion

Linguists [Experiencer] can't help their **appeal** for **terrible examples** [Stimulus].

The six frames

Who must report what, to whom, under which conditions, by when, how often

ACTIONS

Telling

The reporting act: Speaker, Addressee, Message, Topic, Medium

Intentionally_create

Producing the artifact: Creator, Created_entity, Purpose

Conditional_scenario

The condition under which the requirement applies

Time_vector

The deadline: Direction, Landmark_event, Distance

Frequency

How often the act recurs

Artifact

The reported object and its part-whole structure

CONSTRAINTS

All six are unmodified FrameNet frames — no bespoke inventory was created.

REPORT ANATOMY

Three annotation layers

Annotated in INCEpTION

LU

The evoking word, tagged lemma.POS

develop.v

- 1 In order to ensure consistent application of this Article, ESMA shall develop.v draft regulatory technical standards specifying the contracts that are considered to have a direct, substantial and foreseeable effect within the Union or the cases where it is necessary or appropriate to prevent the evasion of any provision of this Regulation as referred to in paragraph 1(a)(v), and the types of indirect contractual arrangements that meet the conditions referred to in the second subparagraph of paragraph 3.
- 2 ESMA shall submit.v those draft regulatory technical standards to the Commission by.prep 2012-09-30.
- 3 Power is delegated to the Commission to adopt the regulatory technical standards referred to in the first subparagraph in accordance with Articles 10 to 14 of Regulation (EU) No 1095/2010

Three annotation layers

Annotated in INCEpTION

LU

The evoking word, tagged lemma.POS

develop.v

Frame

Same span as the LU, labelled with the frame name

Intentionally_create

- develop.v

Intentionally_create
- 1 In order to ensure consistent application of this Article, ESMA shall

develop

 draft regulatory technical standards specifying the contracts that are considered to have a direct, substantial and foreseeable effect within the Union or the cases where it is necessary or appropriate to prevent the evasion of any provision of this Regulation as referred to in paragraph 1(a)(v), and the types of indirect contractual arrangements that meet the conditions referred to in the second subparagraph of paragraph 3.
 - 2 ESMA shall

submit.v

Telling

 those draft regulatory technical standards to the Commission

by

by.prep

Time_vector

 2012-09-30.
 - 3 Power is delegated to the Commission to adopt the regulatory technical standards referred to in the first subparagraph in accordance with Articles 10 to 14 of Regulation (EU) No 1095/2010

Three annotation layers

Annotated in INCEpTION

LU

The evoking word, tagged lemma.POS

develop.v

Frame

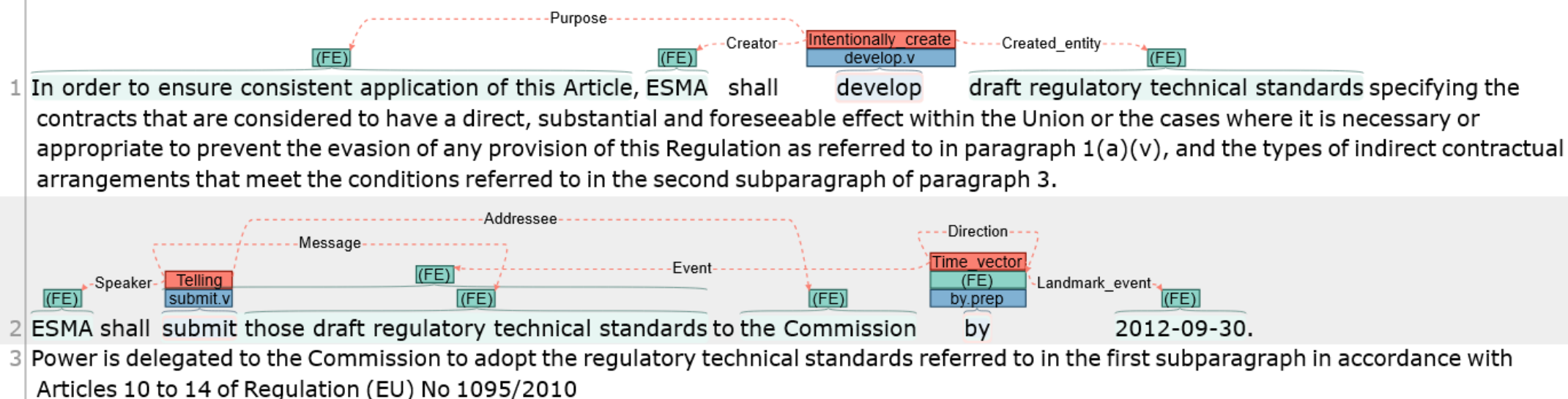
Same span as the LU, labelled with the frame name

Intentionally_create

FE

One span per argument, linked to a role

Creator, Created_entity



Key annotation conventions

Spans

Contiguous, but may nest. A span filling roles in two frames is created once and linked twice.

Coordination

Coordinated fillers get one span and one link per conjunct, not one span over the whole phrase.

Time_vector

A bare date or named reference point is a `Landmark_event`; an explicit numeral + unit is a `Distance`.

Frequency

The Event is always linked to the governing predicate or event-denoting noun.

Implicit agents

In an agentless passive, the party role is left unlinked rather than inferred.

The guidelines are published with the corpus so a second annotator can reproduce the scheme.

Inter-annotator agreement

Protocol

The annotator

Background in law and technology, native English speaker, no prior FrameNet experience.

The material

A local INCEpTION project with identical layers and tagsets.

The constraint

No communication between annotators.

19

provisions double-annotated

Agreement is reported separately for the three sub-tasks of frame-semantic parsing, to avoid conflating different sources of disagreement.

Agreement results

The rows form a pipeline: each is conditioned on agreement at the previous stage

Sub-task	Metric	Value	Computed over
Target identification	Jaccard	0.759	66 of 87 positions marked by either annotator
Frame identification	Cohen's κ	1.000	on the 66 positions both marked
Argument identification (exact)	F1	0.607	identical span boundaries
Argument identification (relaxed)	F1	0.824	Dice $\geq$ 0.5
Role classification	Cohen's κ	0.994	on spans both annotators agree exist

Disagreement is almost entirely about where to annotate, not what to call it.

What the disagreement was

18 frame instances added by the second annotator, 3 by the reference annotator

Guideline gaps

Unclear definition of RR

Minor disagreements: coordinated-verb under-segmentation, a competing frame choice, and a missed immediacy adverb.

Roughly half the disagreement traces to two specific, fixable gaps rather than to unreliable categories.

Mapping onto RRMV

Regulatory Reporting Metadata Vocabulary, Release 1.0.0

Clean fits

- Time_vector → InstantSpec / DurationSpec, before / after
- Frequency → hasFrequency (value + unit)
- develop-then-submit → Action.hasNext
- Addresser → :addresser, Addressee → :addressee
- Frame/FE spans → hasTarget with character offsets

Gaps identified

- No property for the communication Medium
- No property for an artifact's Purpose
- No Condition class for conditional applicability
- No part-whole property on ActionResult
- No ResultType for opinion or proposal artifacts

Because the gaps are grounded in real annotated text rather than abstract ontology review, they are concrete proposals for a future revision. RRMV's successor LOMO already closes two of them.

Limitations

Scale

This is a pilot resource.

Guidelines developed on this corpus

Generalization is untested.

Single reference annotator

One agreement round.

Future work

1

**Expand corpus &
extend the agreement
study**

2

**Train and evaluate
extraction**

3

**Scale the ontology
population**

Takeaways

RQ1

FrameNet's existing inventory is expressive enough to capture addresser, addressee, content, condition, deadline and frequency, with no bespoke scheme.

RQ2

Frame- and role-label agreement are near-perfect whenever both annotators identify the same span ($\kappa = 1.000$ and 0.994); the remainder traces to a few fixable gaps.

RQ3

Several frames map cleanly onto RRMV; others expose concrete gaps. Both outcomes are informative.

Guidelines, annotated corpus, RRMV mapping (JSON / CSV) and the worked Turtle example are released under CC BY 4.0

github.com/lecatcel-ops/rr_fn_annotation



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Thank you

Questions and comments welcome

Célestine Juliette Marie Lecat · CIRSFD ALMA-AI, Department of Legal Studies - University of Bologna, Bologna, Italy